Villarrica Homestead Master Plan v5.2

Final drawing set + automation layer + private compound, pool and two-layer philosophy (v5.2) · nature-focused private home · Villarrica, Guairá, Paraguay · 2.0 ha · all yields still untested planning estimates

Goal: Maximum self-sufficiency with lowest practical daily labour through reliable automation and closed loops. Designed for a single operator who lives, trains, makes music, and enjoys nature and swimming on the land. The home compound is private, for us only: no entertaining, no showing off. Every animal feed is grown on the property.

Residents: a 26-year-old man (target 125 kg, building muscle) and one woman; capacity for a second woman later (this would need a nutrient and area rebalance). Blue-shaded rows in tables mark items that are new in v5, v5.1 or v5.2 (v5.2 adds the private compound, lap pool and two-layer philosophy; see section 1c).

Design philosophy (v5.2): two layers, with automation and repairability as the enabler

Layer	What it means here	Decisions it drove
1. Private compound (Zone A): daily enjoyment and recovery	Nature, training, swimming, PC / TV / sport, music, cooking and rest, for a single resident. Private, for us only: no entertaining, no showing off.	Deep 3 m north veranda as the main living room; showpiece soil beds and garden; 25 m lap pool beside the gym patio; detached music room and gym annex; TV / sport lounge, separate PC room, potting room; dog yard; fire pit; tree belts on every side; core 2,058 m² (≈2,400 m² with belts).
2. Productive land (Zones B–F): reliability, then labour, then efficiency	Nothing critical has a single point of failure; every animal feed grown on site; lowest practical daily labour; closed loops.	Freezer room (4 × 300 L) beside the inverter wall; ≥3–4 days battery plus generator (ATS); four go/no-go gates; 1,020 m² expansion reserve; well and pump 30 m+ from animals and humanure; +30% margin on every food (eggs excepted); detached work barn; short one-way flows; four large cook weeks.
Enabler: automation and repairability	Sensors + simple mechanical systems + a spare-parts kit rank above complex robotics. Tie-break: life-critical systems (water, power, freezers, animal welfare) win any conflict between the layers.	Water, power, feed, climate, animal and plant automation limited to what works at 2 ha / 1 operator; manual override on everything; spare-parts store; the pool pump sits on the non-critical bus and is shed first.

What is in this document

Section	Contents
1	Critical numbers (with new v5 rows), go/no-go gates, what changed v1 → v5
1b	Design review v5 / v5.1 / v5.2: what moved and why; what I am happy / not yet happy with
1c	**NEW v5.2: two-layer design philosophy** (private compound vs productive land; rings, sun and wind, noise gradient, hobbies)
2	Property layout and six zones (A–F), site plan, block schedule, walking distances, rationale
3	Private living zone and house concept, house plan, room schedule, phasing; 3b work barn
4–6	Diet and supply, egg decision, animals and feed, plant methods and fertility
7–8	Cook weeks, meals, storage, year calendar, weekly diet, animal feeding, cook schedule
9	**NEW: Automation layer** (priorities, device and method grid, architecture diagram, exclusions, failure modes)
10–11	Labour (with automation effect); water, power (re-run with automation loads), sanitation
12–15	Capital and build order, trials scoreboard, hard rules, risks, legal checklist, closing notes

Items marked “v4 decision” are carried from v4; items marked “v5” are new in this document; items marked “proposal” are suggestions to verify locally. Capital costs are left as TBD because no prices were supplied.

1. Critical numbers (v4 values kept; v5 rows added)

Item	Final figure	Notes
Land / active area	2.0 ha / ~1.66 ha active (16,562 m²)	v5.2: +0.16 ha vs v4 because Zone A (private compound with pool) and Zone C (smelly strip) replace the 1,000 m² hub; solar adds ~200 m²; check against labour
Goat forage	6,500 m² (5 paddocks × 1,300 m²)	Largest single block, rotational
Food crops + orchard	2,650 m² (staples 1,950 + orchard 550 + rain shelters 156)	staples sum 255+245+240+370+120+410+60+250
Duck maize + buffer	1,800 m² (1,500 + 300 buffer)	Drawn at the safe end
Duck pasture + house	800 m² (25 × 32 m) in 3 rotation cells + 25 m² raised house	South-east beside Ponds 3–4 and the barn grain store; ~70 birds on site × ~11 m²
Ponds / duckweed	4 × 500 = 2,000 m² / 200 m²	Tilapia + traíra; one deep covered pond
Zone A / Zone C / work barn	2,058 m² / 424 m² / 125 m²	Zone A: house, garden, pool, dog yard, annex. Zone C: humanure, compost, mushroom, BSF, firewood. Barn: slaughter, workshop, grain, dry goods, spare parts
House	235.2 m² house (24.0 × 9.8) + 72 m² veranda + 49.5 m² detached activity annex (+27 m² patio)	v5.1: music room and gym detached; PC room and potting room added; dry-goods store moved to the barn
Muscovy / goats	18–22 hens + 4–5 drakes / 6 does, 1 buck, 1 wether, kids	Muscovy only; no Khaki
Eggs / milk / fish / meat	~2,200–2,800 / ~1,100 L / ~150 kg fillet / ~175 kg per year	Egg need reset to ~2,300
Cook weeks	4 a year, ~13 weeks (91 days) apart: Feb, May, Aug, Nov	~455 kg, ~550–600 L of frozen cooked food per cycle
Freezers / power	4 × 300 L = 1,200 L / ≥3–4 days autonomy	Freezers ~1,000 kWh a year
Well + pump house	3 × 3 m, south of the compound beside the solar strip (v5.2)	Gate 1; position confirmed only after the yield test
Labour	≤5–6 h/day average outside cook weeks	Target only; log hours in years 1–2; automation expected to reduce chores (section 10)
Private living zone (v5)	Zone A core 2,058 m² + belts ≈ 2,400 m² (private compound)	v5.2; private, for us only; still a concept until surveyed
Lap pool (v5.2)	25 × 4.5 m (112.5 m²), ~150 m³, in Zone A beside the gym patio	Evaporation ~0.5–0.7 m³/day in summer and filtration ~3 kWh/day (estimates); non-critical bus; cover and fence
Automation power budget (v5)	Critical loads only (freezers, pumps, controllers, sensors) with 3–4 day autonomy	Re-run in section 11
Sensor & control nodes (v5)	Node-RED / MQTT / Grafana (or equivalent) dashboard	Phone alerts, trends, fault history
Spare-parts kit (v5)	Mandatory	Pumps, valves, filters, sensors, irrigation fittings, belts, fuses, etc.

Hard go/no-go gates before building (all kept from v4)

- Water rights and a sustained well-yield test (peak need ~7 m³/day in the driest month).
- Confirm maize, cowpea and sunflower can be sown twice a year on the same ground; if not, double those three areas (the expansion reserve exists for this).
- 20 m² trials of tomato/pepper/greens in soil beds and of chia.
- BSF output trial before fixing the duck-maize area.
- v5 addition (check, not a fifth gate): confirm local availability of simple sensors, valves and Node-RED-compatible hardware before committing to the automation design.

What changed

Version	Change
v1 to v2	Hydroponics removed; pacú replaced by tilapia + traíra; heart dropped; cheese 22 to 11 kg; 9 meals; 4 cook weeks; +30% surplus on every food.
v2 to v3	Khaki ducks removed (Muscovy only); +300 m² duck pasture; +300–600 m² maize buffer; full house; water, labour, power, humanure and capital sections added; eggs lowered.
v3 to v4	Egg gap decided; house resized and drawn to scale; property laid out to scale; slaughter and workshop moved to a detached work barn; ducks moved south-east with an 800 m² pasture; well and pump placed; 13-week cook cycle; weekly diet, animal feeding and a balanced cook week added.
v4 to v5	House re-oriented as a private nature-focused home (guest house / formal entertaining removed); private garden buffer; six-zone property structure; full automation layer (water, feed, power, climate, animals, plants, monitoring); practicality filter applied; cook-week flow reinforced with alerts; power and risk sections re-run for automation.
v5 to v5.1	Site re-cut around a real private zone (Zone A) with garden, fire pit, dog yard and annex; smelly uses moved to a strip west of the house; music room and gym detached; PC, TV / sport and potting rooms added; dry-goods store and spare-parts store in the barn; well moved beside the main solar array; expansion reserve under the maize block; design philosophy section added.
v5.1 to v5.2	Two-layer philosophy (private compound for enjoyment; productive land for reliability); compound enlarged to core 2,058 m² (≈2,400 m² with belts) and made explicitly private; 25 × 4.5 m lap pool beside the gym patio; dog yard moved south of the house; Zone C narrowed; solar moved to a south strip and well beside it; pool added to water, power, capital and risk sections.

1b. Design review: what changed in v5, v5.1 and v5.2, and why

Change	Why	Where
Workshop and wet processing in a detached work barn (kept from v4); barn 125 m² with dry-goods and spare-parts stores	Raw meat, blood, offal, fuel, rodents and bulk deliveries stay out of the house; the spare-parts kit has a labelled home.	Barn plan (3b)
v5: Private nature-focused home; full automation layer; practicality filter; alerts on the cook-week flow	One resident's priorities; fewer chores; failures visible early.	Sections 3, 9
v5.1: Site re-cut around a real private zone; smelly strip west of the house; music room and gym detached; potting, TV / sport and PC rooms; dog yard	The v4 hub could not hold house, garden, dogs, fire pit and annex; noise and mess ordered away from sleep and relaxation.	Sections 2, 3
v5.2: Two-layer philosophy: the compound is optimised for private daily enjoyment; the productive land for reliability, nutrient targets, low labour and closed loops	Matches the latest direction: pleasure and recovery in the compound, reliability and efficiency on the land. Life-critical systems still win any conflict (see 1c).	Front page, 1c
v5.2: Compound enlarged: core 2,058 m² + belts ≈ 2,400 m²; explicitly private, for us only (no entertaining, no showing off)	More garden depth, stronger enclosure and privacy; room for the pool without crowding the garden.	Site plan
v5.2: 25 × 4.5 m lap pool in Zone A along the east edge, its south end meeting the gym patio	Daily train → swim → shower → recover flow; the pool is visible from the veranda and the garden.	Sections 2, 3
v5.2: Dog yard moved south of the house beside the mudroom (dog wash); annex shifted south; Zone C narrowed to 8 m; solar moved to a strip south of Zone A; well beside it	Makes room for the pool and garden depth without moving any productive block.	Site plan
v5.2 not adopted: swimming in a farm dam or fish pond	Ponds carry duck manure and fish and are managed as a food system; a dedicated pool keeps swimming clean and the ponds safe.	Section 3
v5.2 kept as is: veranda depth 3.0 m	3 m gives the eave shading in the sun-path logic; a pool-side deck adds the extra outdoor floor instead. Widen only after a sun-path check.	Section 3

Note: the site plan and house are a concept re-cut on assumed wind, slope and plot shape. Scaled construction drawings should only be made after the survey and once the compound size is locked.

What I am happy with

- Two clear layers: a private compound for enjoyment and a productive landscape for reliability. The compound can be generous without weakening any nutrient or labour logic on the land.
- The cook-week flow is still one line (barn → mudroom → processing → freezers → pantry → kitchen), now running one way along the south side of the house.
- Train → swim → shower is a straight line: gym patio opens onto the south end of the pool.
- Goats and ducks stay at opposite ends; smell goes south-west, away from the compound; ponds stay on the low east side and bed runoff never reaches them.

What I am still not fully happy with

- **Active area grew** from ~1.50 ha (v4) to ~1.66 ha (16,562 m²). The compound, pool and smelly strip are the cause. Phase the build and watch labour.
- **Pool upkeep and cost**: it adds water use, filtration power and maintenance (section 11). Use a cover, solar-direct pumping and a simple filtration system.
- The well sits ~20 m from the duck-pasture edge (about 36 m from the duck house, 54 m from compost). Seal the well head and verify with a hydrogeologist and MADES after the yield test.
- The garden still looks north over the goat paddocks; if the survey shows wind from the north, swap goats and the reserve.
- Everything assumes a rectangular 160 × 125 m plot, an east fall, NE summer breeze and southern cold winds. None is surveyed.

1c. Design philosophy (v5.2): two layers

In one sentence: the private compound is for daily enjoyment and recovery; the productive land is for reliability, nutrition and low labour. Pleasure close, chores in the middle, smell and noise at the edges, with sun, wind and slope deciding the rest.

	Layer 1: the private compound (Zone A)	Layer 2: the productive land (Zones B–F)
Optimised for	Nature, training, swimming, PC / TV / sport, music, cooking, rest. Private, for us only: no entertaining, no showing off.	Reliability, nutrient targets, lowest practical labour, closed loops, +30% margins, go/no-go gates.
Success looks like	You want to be there; every day feels good; nothing smelly, loud or muddy intrudes.	The freezers are full, the animals are fed and safe, hours stay under 5–6 a day, and nothing critical has a single point of failure.
Decision rule	Comfort and beauty win, within budget and upkeep you can sustain.	Reliability first, then labour, then everything else.

Tie-break: *where the layers conflict, life-critical systems (water, power, freezers, animal welfare) win; on everything else the compound wins on how it feels and the land wins on how it works. The pool, for example, sits on the non-critical power bus and is shed first.*

Property philosophy: four rings around the compound

Ring	What lives there	Rule that governs it
1. Home and joy (Zone A, ≈2,400 m² with belts)	House, 3 m north veranda, showpiece soil beds, lawn + fruit trees, fire pit, lap pool, dog yard, detached music + gym annex	Nothing smelly, muddy or loud. Tree belts on every side. The view is north over the garden and the grazing paddocks.
2. Daily chores (under 50 m)	Goat shed 47 m, ducks 29 m, BSF 34 m, compost 37 m, mushroom room 34 m, barn 16 m, well and solar 23–25 m	Every daily walk is under a minute; flows are one-way; paths stay dry; daily chores are automated first.
3. Production (50–90 m)	Food block, orchard, rain shelters, expansion reserve, duck maize, goat paddocks	Largest areas sit furthest, are easiest to fence and rotate, and hold crops that tolerate a visit every few days.
4. Water and wild edge	Ponds on the low east side, windbreaks south and east, well and solar strip south	Runoff runs away from ponds; the well keeps its distance from animals, humanure and compost; the edge is also habitat.

The five forces that placed everything

- **Sun (southern hemisphere):** veranda, garden and pool face north for winter sun; the 3 m eave shades summer sun. PC room and kitchen sit on the shaded south side.
- **Wind:** NE summer breeze carries smell south-west, so humanure, compost, BSF and firewood are in a strip west of the compound, with a hedge between. The house shelters the garden from cold south wind.
- **Slope and water:** assumed fall to the east, so ponds are east, beds are west, and runoff never meets the fish. Pool water never goes to ponds or beds.
- **Clean and dirty:** clean side is north (garden, veranda, pool). Dirty side is south, east and west edges: road, barn, ducks, Zone C. Trucks and slaughter never pass the veranda.
- **Reliability:** well, solar strip, generator and freezers are within about 25 m of the mudroom door; a failure is a short walk, not a trek.

House philosophy: rooms ordered by noise and mess

Part of the house	Rooms	Why it is there
West wing: hobbies + social	Potting + seed room (door to garden), living + dining + TV / sport wall, shaded PC + media room, kitchen	Evenings: sport, films, Netflix, gaming, seedlings. Casual, resident-only spaces. The potting room is the one room allowed to be muddy.
Middle: sleep	Main bath (sound buffer), master + walk-in + en-suite, bedroom 2	The bath sits between the TV and the master so late matches do not wake anyone. Bedroom 2 is a flexible partner / work room.
East wing: work	Mudroom + inverter wall, processing + storage, freezer room, pantry	Cook-week flow runs east to west into the kitchen, with no backtracking.
Detached: loud + wet	Music room + gym annex with covered patio, 3 m east of the house, opening to the pool	Dropped weights, guitar and amps never share a wall with bedrooms or the TV room. Train, swim, shower, recover in a line.
Detached: messy	Work barn: slaughter, workshop, grain, dry goods, spare parts, chill room	Blood, fuel, rodents and bulk deliveries stay out of the compound.

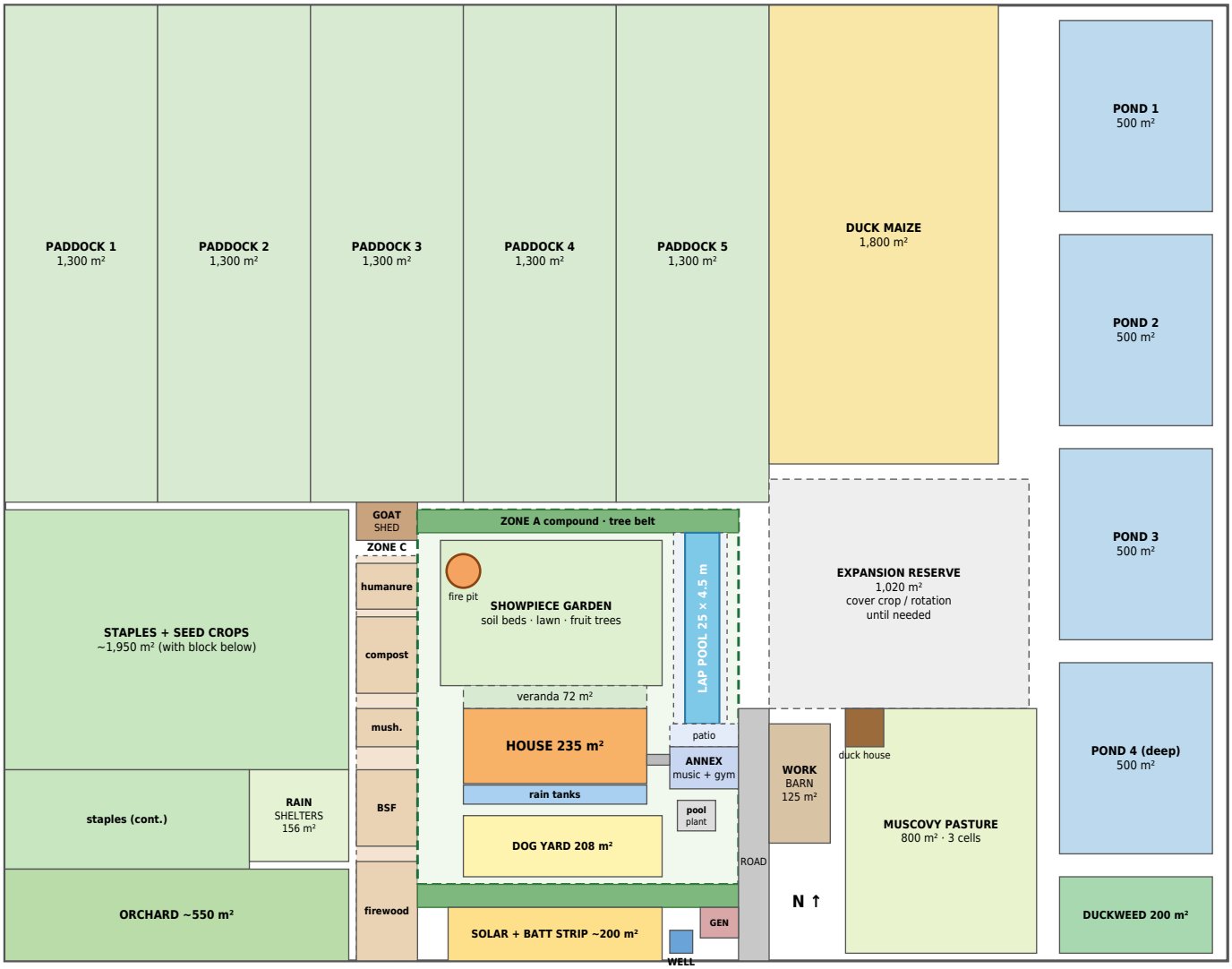
Your hobbies, and where each one lives

You like	Where it lives
Gym	Annex gym 24.8 m², rubber floor, rig, big doors onto the covered patio and the pool.
Swimming	25 × 4.5 m lap pool along the east edge of the compound, directly beside the gym patio; cover, solar-direct pump, shower and towels at the plant room.
PC, TV, movies, sport, Netflix	Living-room TV wall at the west end, dimmable blinds, wired network; separate shaded PC + media room (17.1 m²); optional projector screen on the veranda.
Nature, growing things, soil	Showpiece soil beds in front of the veranda (compost vs mulch vs biochar test beds, soil journal, sensors on the dashboard), fruit trees in the garden, potting + seed room opening onto the beds.
Dogs and animals	Fenced dog yard (208 m²) south of the house beside the mudroom dog wash, shaded rest spot; gate to the garden; apart from ducks and goats until trained.
Guitar	Music room in the annex, thick walls and a sealed door; fire-pit corner at the garden's edge for evenings.

2. Property layout and zones (v5.2)

Six zones. The private compound (Zone A) is larger and more enclosed than in v5.1; the productive blocks are unchanged. Goats stay north-west and ducks south-east, with the whole site between them.

Zone	Contents	Notes
A – Private compound (core 2,058 m², ≈2,400 m² with belts)	House + deep north veranda, showpiece garden, lawn, fruit trees, fire pit, 25 m lap pool, dog yard, activity annex	North tree belt (toward goats), hedge on the Zone C side, south and road-side belts; private, quiet, sunny
B – Utilities & resilience	Well + pump house, solar + battery strip, generator (ATS), rain-tank bank at the house	South of the compound; 23–25 m from the mudroom door
C – Smelly-uses strip + work barn	Humanure, compost, mushroom room, BSF shed, firewood (strip x 46–54 m); slaughter / processing, grain, workshop, dry goods, spare parts (barn)	West and south-west, downwind of the assumed NE breeze; barn on the road side
D – Food crops + orchard + rain shelters	Staples, orchard, shelters	West; bed runoff never reaches ponds
E – Duck system	800 m² pasture (25 × 32 m) + house + Ponds 3–4	South-east, crosswind; duck house 2–10 m from the barn grain store
F – Goat forage	5 rotational paddocks + shed	North-west, opposite end from ducks



Site plan to scale (1 m = 3.2 pt; land 160 × 125 m = 20,000 m²). North is up. Concept only: assumed fall east, NE summer breeze, southern cold winds, windbreak on the south and east edges. Dashed green = Zone A. Named blocks total 17,787 m² (88.9%); the remaining 2,213 m² is planted belts, windbreak, dikes, paths and road.

Block schedule (areas computed from the drawn rectangles)

Block	x, y (m)	Size (m)	Area m²	Why here
Goat forage + browse (5 paddocks)	0, 0	100 × 65	6,500	North-west; goat shed at (46, 65) with a 3 m service alley
Duck maize + buffer	100, 0	30 × 60	1,800	Next to forage; grain goes to ducks and fish
Expansion reserve	100, 62	34 × 30	1,020	Under the maize block it would extend; cover crop until needed
Ponds 1–4 (500 each)	138, 2 / 30 / 58 / 86	20 × 25	2,000	Column on the assumed low side; Pond 4 deep and covered
Duckweed	138, 114	20 × 10	200	Ponds feed it; ducks and fish eat it
Food crops + orchard block	0, 66	45 × 59	2,655	West; staples, shelters (32, 100), orchard (0, 113)
Zone C strip	46, 72	8 × 53	424	Humanure, compost, mushroom, BSF, firewood
Zone A private compound (core)	54, 66	42 × 49	2,058	House (60, 92), garden (57, 70), pool (89, 69), dog yard (60, 106), annex (87, 97)
Muscovy pasture (3 cells)	110, 92	25 × 32	800	South-east; duck house at NW corner; gated path to Ponds 3–4
Work barn	100, 94	8 × 15.6	125	Road side; grain store 2–10 m from the duck house
Solar + battery strip	58, 118	28 × 7	196	South of the compound, unshaded; generator shed (91, 118)
Well + pump house	87, 121	3 × 3	9	30 m+ from duck house and goat shed, ~50 m from compost and humanure; final spot after yield test
Named blocks total			17,787	88.9% of 20,000 m²; remainder 2,213 m²

Walking distances (from the mudroom door, straight line)

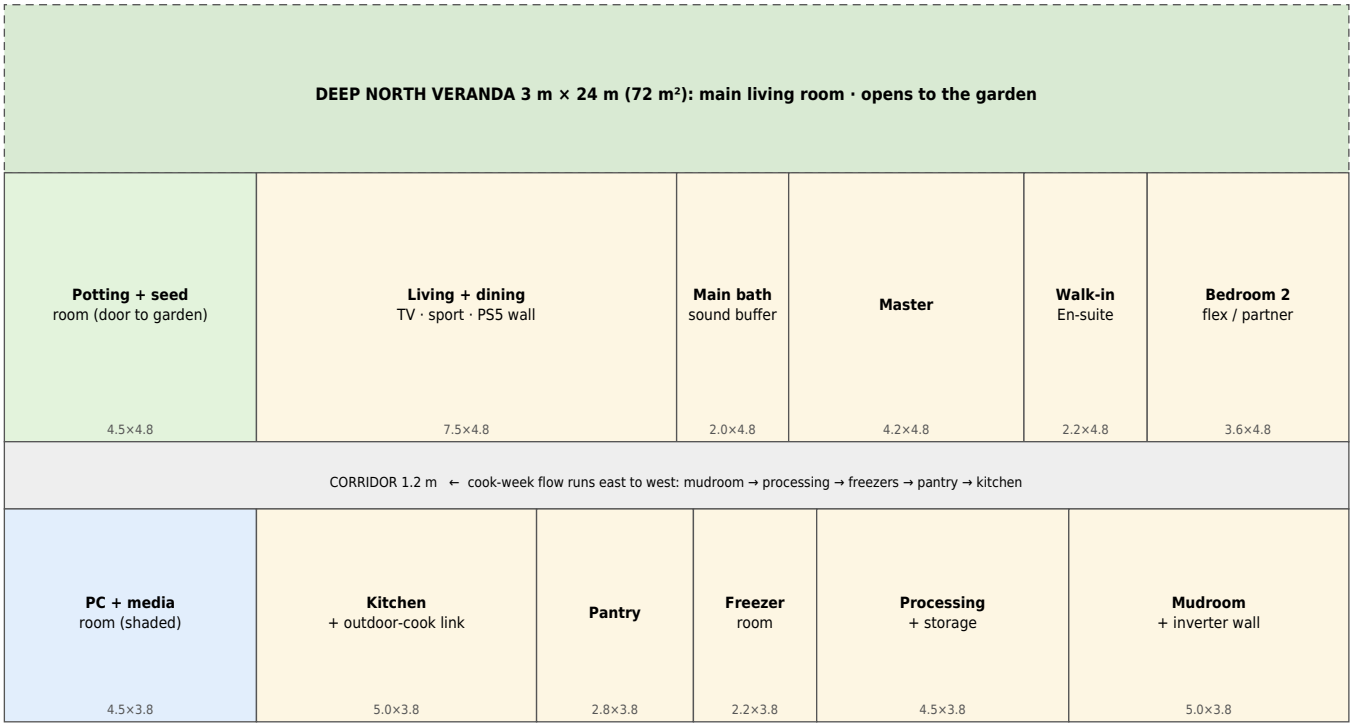
Chore point	Metres	Walk at 1.3 m/s	Chore point	Metres	Walk
Goat shed + milking	47	0.6 min	Orchard (centre)	64	0.8 min
BSF shed	34	0.4 min	Pond 4 (deep)	64	0.8 min
Duck house	29	0.4 min	Duckweed	67	0.9 min
Work barn (door)	16	0.2 min	Solar strip	25	0.3 min
Compost	37	0.5 min	Well + pump house	23	0.3 min
Humanure	42	0.5 min	Showpiece garden	24	0.3 min
Mushroom room	34	0.4 min	Pool (centre)	20	0.3 min
Rain shelters	46	0.6 min	Fire pit / dog yard	35 / 15	0.4 / 0.2 min

Property design rationale and assumptions

- Compound: Zone A is a 42 × 49 m rectangle (2,058 m²) plus ~330 m² of outside belts, ≈2,400 m². The house sits low in it so the veranda looks north across 19 m of garden, with the pool along the east edge in view.
- Privacy: tree belts on every side, a hedge toward Zone C, road and barn behind the house rather than in front of the veranda; no entertaining spaces; one gate.
- Reliability: well, solar strip, generator and freezers all within ~25 m of the mudroom door; the 1,020 m² reserve sits under the maize block and is sized to the exact area needed if double sowing fails.
- Smell and wind: humanure, compost, mushroom, BSF and firewood sit in a strip west of the compound; the NE summer breeze carries smell south-west, away from Zone A.
- Ducks: 800 m² pasture in the south-east with a gated path to Ponds 3–4; duck house beside the barn grain store. Goats are in the north-west, so the whole site lies between them.
- Well: south beside the solar strip; 30 m+ from duck house and goat shed, ~50 m from compost and humanure, ~20 m from the pasture edge (proposal: seal the well head; verify with a hydrogeologist and MADES).
- Pond runoff rule: shelters and beds in the west, ponds in the east: bed runoff never reaches ponds. Pool drain water never goes to ponds, beds or the well.
- Assumptions to verify on the survey: slope falls east; NE summer breeze; cold winds from the south; plot roughly 160 × 125 m and rectangular. If it differs, keep block areas and re-place them with the same rules.

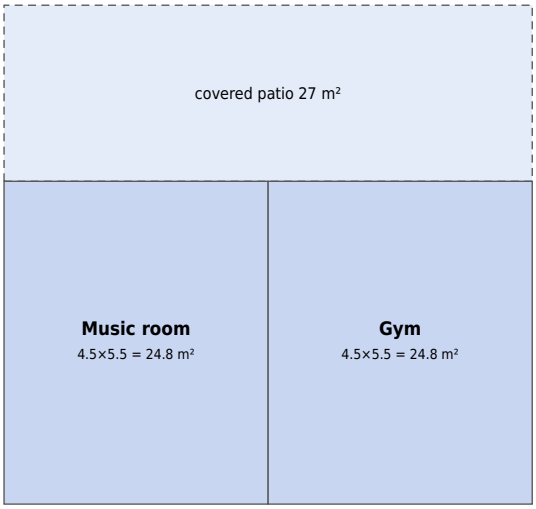
3. House concept and floor plan (v5.2)

A **nature-focused, low-maintenance private home**: 24.0 × 9.8 m main house (235.2 m²) + 3 m north veranda (72 m²) + detached activity annex (49.5 m² + 27 m² covered patio) + 25 m lap pool beside it. The guest house and formal entertaining spaces are removed; rooms are ordered by noise and mess (see 1c).



Concept diagram, single storey, 1 m = 21 pt. North row west to east: potting + seed room, living + dining + TV, main bath (sound buffer), master with walk-in and en-suite, bedroom 2. South row (opens to the 1.2 m corridor): PC + media room, kitchen, pantry, freezer room, processing + storage, mudroom with inverter wall. Scaled construction drawings still to be made after the survey.

Detached activity annex (music + gym)



9 × 5.5 m, 3 m east of the house (x 87–96, y 97 on the site plan), directly south of the lap pool and joined to the house by a covered walkway so there is no wet walk to the gym.

Music room 24.8 m²: double-leaf sealed door, dense walls, floating floor option, acoustic panels; far from the bedrooms. **Gym 24.8 m²:** rubber floor, rig, mirror wall, two big doors onto the 27 m² covered patio, whose north edge meets the south end of the lap pool, with an outdoor shower between.

Power and network come from the house via the same walkway trench; the annex is on the non-critical bus (shed first).

Lap pool (v5.2)

Item	Plan
Size and position	25.0 × 4.5 m (112.5 m²), long axis north–south along the east edge of Zone A (x 89–93.5, y 69–94 on the site plan); south end meets the gym patio; 1.5 m deck each side; depth 1.2–1.5 m (~150 m³). Visible from the veranda and the garden.
Daily flow	Gym → patio → pool → outdoor shower → house: train, swim, shower, recover in one line, with no wet walk through the house.
Water	Initial fill ~150 m³ from tanks or well in the wet season; summer evaporation estimated ~0.5–0.7 m³/day (about 8–10% of the 7 m³/day peak); cover when not in use. Added to the water balance (section 11).
Power	Filtration pump ~0.5 kW × ~6 h ≈ 3 kWh/day (estimate), run solar-direct in daytime on the non-critical bus; first load shed in any power shortfall.
Treatment and plant	Simple cartridge or sand filter with salt chlorination or similar low-chemical system (proposal); 5 × 4 m plant room at (88, 104) with the pump, filter and controls on the dashboard.
Heating and season	No heating in the baseline; a solar cover extends the season. Villarrica winters are cool, so expect a swim season of roughly October to April (assumption).
Drain and backwash	To a soak pit near Zone B; never to ponds, beds or the well. Pool water and chlorine stay out of the food system.

Item	Plan
Safety	Fence with self-closing gate on the dog side, pool alarm, cover, non-slip deck, dog ramp; when swimming alone, send a phone check-in or use the camera and alert on the dashboard.
Cost	TBD (quote locally; a swim-spa or shorter pool is the fallback if budget or water is short).

Swimming in a farm dam or fish pond is not adopted: the ponds carry duck manure and fish and are managed as a food system.

Room schedule

Room	Module size (m)	Area m²	Role / placement reason
Potting + seed room	4.5 × 4.8	21.6	West end, door to the garden; sink, bench, seed starting, tools; the one muddy room
Living + dining + TV / sport wall	7.5 × 4.8	36.0	Opens to veranda; TV wall, blinds for daytime matches, speakers, wired network
Main bath	2.0 × 4.8	9.6	Sound buffer between TV room and master
Master bedroom	4.2 × 4.8	20.2	Quiet middle, garden view; walk-in and en-suite attached
Walk-in / En-suite	2.2 × 2.4 each	5.3 + 5.3	Doors from master only
Bedroom 2	3.6 × 4.8	17.3	Flexible partner / work room at the east end
PC + media room	4.5 × 3.8	17.1	South side (no glare), dimmable light, big desk; gaming and sport
Kitchen	5.0 × 3.8	19.0	South of living, visual link to veranda; outdoor-cook option on the veranda
Pantry (with cleaning closet)	2.8 × 3.8	10.6	Between kitchen and freezers
Freezer room	2.2 × 3.8	8.4	4 × 300 L chest freezers, own ventilation, temperature alarms, beside the inverter wall
Processing + storage	4.5 × 3.8	17.1	Vacuum sealing, portioning; harvest enters from the mudroom
Utility / mudroom	5.0 × 3.8	19.0	East entrance by the road; boots, laundry, dirty shower, inverter wall, controller node, dog wash
Corridor	24.0 × 1.2	28.8	Every room opens onto it
Total house	24.0 × 9.8	235.2	Rooms 206.4 + corridor 28.8; dry-goods store and office removed (barn / PC room)
Veranda	24.0 × 3.0	72.0	Primary living room: hammock, sofa, outdoor dining, cook-week overflow
Annex: music + gym	9.0 × 5.5	49.5	Detached; plus 27 m² covered patio

Phasing and cut list

Phase	Build	Area	Notes
Core (build first)	Living, bath, master suite, bedroom 2, kitchen, pantry, freezer room, processing, mudroom, corridor (x = 4.5 to 24.0 m of the plan)	191.1 m² + veranda 58.5 m²	Runs the whole cook-week system.
West wing	Potting + seed room, PC + media room	44.1 m² + veranda 13.5 m²	Can follow once the core stands; a desk in the living room covers the gap.
Activity annex	Music room + gym + patio	49.5 m² + 27 m²	Resident priority: build early if budget allows.
Easiest cuts	PC + media room (use bedroom 2), then the covered patio		Music and gym are no longer cuts.

- High ceilings or roof vents for stack-effect cooling; wide eaves; cross-ventilation (windows on opposite faces of every room).
- Rainwater first-flush into the rain-tank bank along the south wall; size tanks from the monthly water balance.
- Outdoor shower by the mudroom and annex; covered veranda doubles as cook-week overflow workspace.
- Simple, durable materials: concrete floor, block or brick walls, metal roof, solar-ready (inverter wall placed in the mudroom).
- Veranda faces north: winter sun warms it, the 3 m eave shades summer sun (proposal, verify with a sun-path diagram).

3b. Work barn (8.0 × 15.6 m = 125 m²)

Room	Size	Function
Wet processing + slaughter	8.0 × 5.0 m = 40 m²	Slaughter, scald, butcher, fish cleaning; sloped washable slab, floor drains, hot water; livestock and carcasses use the west door; waste leaves by the south door to BSF totes
Workshop	4.0 × 4.0 m = 16 m²	Tool wall, bench, vice, repairs
Grain store	4.0 × 4.0 m = 16 m²	Four rodent-proof bins for ~1,260 kg of duck maize (~1.7 m³), east door to the ducks; short enclosed augers or gravity hoppers
Dry-goods store (moved from the house)	4.0 × 3.6 m = 14.4 m²	Sealed jars and bins for maize, cowpea, peanuts, seeds, chia, honey, dried mushrooms
Spare-parts store (new)	4.0 × 3.6 m = 14.4 m²	Mandatory spare-parts kit: pumps, valves, filters, sensors, fittings, belts, fuses; labelled bins and inventory list
Chill room	4.0 × 3.0 m = 12 m²	Carcasses chill before butchering; battery-backed with temperature alarm
Wash + PPE	4.0 × 3.0 m = 12 m²	Boots, aprons and hand wash before anyone walks to the house

- Waste route (proposal): blood, offal and fish frames to BSF; bones to bone meal; wash water to a soak pit away from the well and ponds.
- Siting: east of the road at x 100–108, y 94–110; duck house 2–10 m east, BSF shed about 50 m west.
- Legal: SENACSA slaughter rules decide what approval the slaughter room needs before use.

4. Diet, targets and supply

Household targets per day (man + one woman): about 8,800 kcal and 445 g protein. Man about 4,400 kcal and 245 g; each woman about 2,200 kcal and 100 g. Table error is about 20%. Rule: freeze as many cooked meals as possible; fresh harvest only for the weekly basket.

Food (household per year)	Need	Food	Need
Cassava / sweet potato / maize	179 / 241 / 102 kg	Mushrooms / honey / chia	55 / 15 / 15 kg
Cowpea / peanuts / sunflower seeds	99 / 22 / 22 kg	Greens, tomato, pepper	329 kg
Eggs	2,920 (v3) → ~2,300 (v4 decision)	Pumpkin / fruit (edible) / avocado flesh	113 / 186 / 57 kg
Milk (drink, yogurt, kefir)	730 L	Goat cheese	11 kg (~110 L milk)
Fish fillet (tilapia)	117 kg	Meat (duck, goat)	128 kg

Heart was dropped (29 kg a year cannot be produced; extra meat replaces it). Cheese was cut from 22 to 11 kg because calcium stayed well above target. Women's iron stays about 35% above target.

Supply versus need (+30% rule; two sowings a year assumed for maize, cowpea, sunflower)

Food	Area / count	Expected edible	Need	Margin
Cassava	255 m²	~248 kg	179 kg	+38%
Sweet potato	245 m²	~313 kg	241 kg	+30%
Maize (people)	240 m²	~143 kg	102 kg	+40%
Cowpea	370 m²	~137 kg	99 kg	+38%
Peanuts	120 m²	~30 kg	22 kg	+36%
Sunflower seed	410 m²	~32 kg	22 kg	+43%
Pumpkin	60 m²	~145 kg	113 kg	+28%
Chia (+ perilla)	250 m²	~20 kg	15 kg	+33%
Cooked greens	~70 m²	~125 kg	95 kg	+30%
Salad + herbs	~25 m²	~78 kg	60 kg	+30%
Tomato + pepper	~60 m²	~195 kg	150 kg	+30%
Orchard (mature)	~550 m²	~290 kg fruit, ~78 kg avocado	186 / 57 kg	+55% / +37%
Eggs (Muscovy)	18–22 hens	~2,200–2,800	2,920 / ~2,300	−25 to −4% / −4 to +22%
Milk + cheese	6 does	~1,100 L	~840 L	+31%
Fish fillet	4 ponds, 2,000 m²	~150 kg	117 kg	+28%
Meat	Muscovy + kids	~175 kg	128 kg	+37%
Mushrooms	20 m² room	~70 kg	55 kg	+27%
Honey	3 hives + 1 stingless	~20 kg	15 kg	+33%

Orchard: 8 banana mats, 3 citrus, 3 guava, 2 avocado (cold-hardy) on ~325 m² of the 550 m² block; the rest holds papaya, mango, pitanga, mulberry and pasture. Avocado yields from year 5–7, citrus and guava from year 3–5. Banana and frozen guava cover the Sep–Nov fruit gap.

Egg decision (v4) and nutrient gaps

Each egg is about 6 g protein and 70 kcal. All 2,920 eggs = ~50 g protein a day (11% of 445 g). A shortfall at the low end (720 eggs) is ~12 g protein (2.7%) and ~140 kcal (1.6%) a day, and meat has a +37% margin. Choline and B12 also come from meat, fish and milk (estimate, verify in the nutrient re-run). **Decision:** reset the egg need to ~2,300 a year by binding sopa, chipa and fish cakes with cassava starch or sweet-potato mash; the 30% surplus rule is relaxed for eggs only. **Fallback:** if year 1 gives under 2,000 eggs, add a separate small Khaki flock as in v2 (20 hens, own 40 m² house). Muscovy lay about 100–150 eggs per hen and the same hens raise ~100 table ducks, so expect the low end. Automated egg collection is deliberately excluded (section 9).

Gaps closed from outside the farm (by design)

- Vitamin D: midday sun, sun-dried mushroom gills, eggs, fish. Test 25-OH vitamin D in winter and supplement if low.
- EPA + DHA: about half of target from food; algae oil ~250 mg/day for the women.
- Iodine: ~5 g/day iodized salt (non-negotiable).
- ALA: ~80% of the man's intake is from chia. Chia is untested, so perilla is the backup; if both fail, ALA falls below target.
- Folate: folic acid supplement if pregnancy is planned. Fibre is high (83 g for the man): build up beans gradually.
- Iron, vitamin A, selenium for the man can run high: ferritin test once, no liver, avoid extra supplements with these.
- Fibre, protein and calories remain the priority. Re-run the nutrient balance after the first full year of egg and yield data.

5. Animals and feed

Animal	Number	Housing	Day range	Notes (v5 automation in blue rows of section 9)
Muscovy ducks (dual-purpose)	18–22 hens + 4–5 drakes	House 25 m² (5 × 5) inside the pasture: raised floor, straw, nest boxes, predator-proof; automatic coop door, locked every night	Own 800 m² pasture (25 × 32 m, 3 rotation cells) with a gated path to the ponds	Natural mothers, no incubator; never mix with other duck breeds
Dairy goats (Anglo-Nubian cross)	6 does, 1 buck, 1 wether, kids	Shed 40 m² (~2 m²/goat) raised slatted floor, dry bedding; buck pen 20 m²; automatic waterer + mineral feeder	5 rotational paddocks, shade, tannin browse; app/timer-controlled gates	Buck penned apart except for breeding
Tilapia + traíra	4 ponds of 500 m²	One deep pond, covered for winter	Duck path along ponds	Self-breeding; traíra limit overbreeding
Black soldier fly	~20 totes	Shaded shed ~60 m² with temperature/humidity sensors	Self-harvest ramps	Key protein for ducks and fish
Bees	3 hives + 1 stingless box	Orchard edge	Farm flowers, sunflower	Honey + pollination, very low labour
Dogs (v5.1)	1–2 companion dogs; calm livestock-guardian dog later	208 m² fenced yard south of the house beside the mudroom dog wash, gate to the garden	Garden and yard only; kept apart from ducks and goats until trained	Fed from offal, fish frames and surplus eggs plus a bought supplement (new external input)

Where the ducks stay (v4 decision)

- Night: in the 5 × 5 m raised house at the NW corner of the pasture (110, 92 on the site plan), 2–10 m from the barn grain store, locked at dusk (now by automatic coop door with a manual override and a camera check) against dogs, foxes and birds of prey.
- Day: the 25 × 32 m pasture, three cells of ~267 m² so each rests two weeks after use; hens with ducklings can have their own cell.
- Ponds: ducks reach Ponds 3–4 through a gated path along the pond edge, not all day, so droppings fertilise the water without turning the ponds to mud.
- Feed: maize from the work-barn grain store 10 m away (timed or gravity-drop feeders), BSF larvae, duckweed, fallen fruit and kitchen scraps. Muscovy are kept apart from other duck breeds and from the goats by distance and a solid fence.

Breeding rhythm

- Muscovy: four hatching rounds a year (~25 ducklings each), timed so ducks reach slaughter weight (12–14 weeks) at each cook week; about 100 table ducks. Incubation ~35 days, so set eggs ~5 weeks before each hatch. Keep the best hens as mothers.
- Replacement stock: keep ~4–7 of the best young hens a year; swap drakes with a neighbour every 2–3 years.
- Goats: two groups of 3 does. Group A bred Mar–Apr (kids Aug–Sep); group B Sep–Oct (kids Feb–Mar). At least 3 does in milk all year; dry off ~2 months before kidding. About 7 wether kids a year raised to ~6 months and slaughtered at the Feb and Aug cook weeks. Keep one doe kid a year; swap bucks every 2–3 years.
- Tilapia: breed freely; seine at cook week, take the large fish, leave the small ones as breeders.

Care routines

- Goats (daily, non-negotiable): milk; heat-stress check in summer; shade and water (level-sensor waterer); weekly FAMACHA parasite check in summer; rotate paddocks (gate automation); tannin browse (moringa); hooves every 3 months; mineral block + salt always (automatic mineral feeder).
- Ducks: collect eggs by hand every 1–2 days; clean water daily; move runs weekly; night lock-up; 14 h of light at dawn in winter if laying drops sharply.
- Fish: weekly water and feeder check; cover or move fish if water falls below 20 °C. BSF: feed twice a week, collect larvae every 2 days (self-harvest totes). Bees: periodic swarm and disease check.
- Health and predators: night lock-up mandatory; solid fence between goat and duck areas; FAMACHA + selective deworming only; quarantine new animals 2–3 weeks; vet kit; predators are dogs, foxes, birds of prey; keep BSF and manure away from the house.

Dogs (v5.1)

- Dogs are family and also the main duck predator: night lock-up for ducks stays mandatory, and the dog yard is fenced away from the pasture.
- Train on leash around goats and ducks for months before any free access; a livestock-guardian dog is bonded to the goats from puppyhood.
- Vaccines, deworming and vet kit join the external-inputs list; offal and fish frames go to BSF first, dogs get only what the farm has spare.

Feed grown on the property

Feed	Need	Supply	Status
Goat forage + browse (napier, moringa, pigeon pea, mulberry, sweet-potato vines, wilted cassava leaf)	~6,000 kg DM/yr	6,500 m² × 1.2 ≈ 7,800 kg	Sufficient

Feed	Need	Supply	Status
Duck grain + protein	~1,200–1,500 kg/yr	1,800 m² maize × 0.7 = ~1,260 kg (1,500 m²: ~1,050) + BSF + duckweed + orchard forage + scraps	Tightest; BSF/duckweed must cover 0–450 kg
Fish feed	Natural + supplement	Duckweed 200 m² (~8,000 kg fresh), BSF, duck manure, second-grade roots	Adequate
Calcium (hens, goats)	Small	Crushed eggshell + bone meal	Closed loop

6. Plant methods, fertility and protection

Crop	m²	Method	Plant / harvest
Cassava	255	Soil rows, 4 blocks	Dig one block per cook week; the rest stays in the ground
Sweet potato	245	3 batches	Harvest Jan, Mar–Apr, Jun; cure and store 3–4 months
Maize (people)	240	2 halves	Sow Sep–Oct and Feb; dry; confirm double sowing
Cowpea	370	2 halves, nitrogen fixer	Sow Sep–Oct and Feb–Mar; dry
Peanuts	120	Sandy-soil rows	Sow Oct–Nov, harvest Mar–Apr
Sunflower	410	2 halves, bee forage	Sow Aug–Sep and Jan–Feb; harvest Dec–Jan and May–Jun
Pumpkin	60	Fence/trellis with compost	Plant Sep and Jan; harvest Dec–Jan and Apr; stores 3–6 months
Chia + perilla	250	Soil rows	Once a year, Apr–May; trial first
Cooked greens (kale winter, amaranth summer)	~70	Compost raised beds, rain cover, soil sensors	One bulk cut at cook week
Salad, herbs, scallion	~25	Raised beds, rain cover, drip	Weekly basket
Tomato + pepper	~60	Beds or buckets under rain shelter, drip + fertigation	Pick, freeze whole; sauce at cook week
Orchard	~550	Tree basins with mulch, drip	Fresh weekly; surplus frozen as compote; full output years 3–7
Mushrooms	20 room	Bags on shelves, dark humid room with climate sensors	New bags every 2 weeks
Duck pasture / maize buffer	+300 / +300–600	Grass-legume rotation / soil rows	Muscovy run / grain safety margin

Fertility with almost no bought inputs

- Sources: goat and duck manure, BSF frass, pond water for irrigation, cowpea and pigeon pea as cover and nitrogen crops, bone meal, wood ash, mulch, biochar, composted humanure.
- Sandy-soil protocol: compost 5–10 cm yearly, biochar 1–2 kg/m² yearly, mulch 5–10 cm, drip irrigation, yearly soil test; one-time lime + rock phosphate if the test requires it. Expect lower yields in years 1–3.
- Never send bed runoff into the fish ponds. Pond water is fine on beds.
- v5: fertigation dosing pumps are driven by the soil EC/pH sensor network but stay small; compost, manure and mulch remain the main fertility.

Crop protection

- Open-sided rain shelters with strong airflow for tomato, pepper, greens (vents + temperature/humidity sensors); mulch, resistant varieties, copper and biofungicide when humidity drives outbreaks.
- Rotation and cover crops (cowpea, pigeon pea) to break pest cycles; manual and biological control before chemicals. No complex AI pest systems (excluded).
- Check beds weekly; sensors and the weather-station feed now flag humidity and moisture problems between visits. Frost cloth (automated shade / frost-cloth on sensitive beds) and frost-tolerant fruit varieties for winter.

7. Cook weeks, meals and storage

Four cook weeks a year (Feb, May, Aug, Nov), about 13 weeks (91 days) apart. The household eats ~5 kg of frozen cooked food a day: ~455 kg or ~550–600 L of flat bags per cycle. Eat oldest first; label every bag with meal and date.

Slot	Meal	v2/v3 per 84 d	v4 per 91 d	Reheat
Breakfast	Sopa paraguaya (maize, cheese, onion, egg, milk)	42	46	Oven 15 min
Breakfast	Mushroom-egg tortilla squares (+ moringa)	42	45	Pan 5 min
Lunch	Goat-bean-greens stew with cassava, pumpkin	36	39	Simmer 15
Lunch	Duck locro (maize, beans, pumpkin)	24	26	Simmer 15
Lunch	Duck or goat curry with sweet potato, moringa, chilli, tomato	24	26	Simmer 15
Dinner	Baked tilapia, tomato-pepper sauce, sweet-potato mash	48	52	Oven 20
Dinner	Tilapia + sweet-potato fish cakes	36	39	Pan 8
Dinner (v3)	Duck and pumpkin hash with eggs	TBD	TBD (carve from locro/curry)	Pan
Snack	Chipa (cassava starch, cheese, egg)	42	46	Oven 10
Snack	Peanut, sunflower, chia, honey, banana bars	42	45	Thaw 10

v2/v3 quantities (336 portions) assumed 84 days; v4 scales them to the 91-day cycle (364 portions, ~1.25 kg each). Flavour bases (tomato-pepper sauce, curry paste, moringa-peanut pesto) are made at cook week with bone broth. Fresh weekly basket: salad, herbs, scallion, moringa, chillies, avocado, citrus, guava, plus daily milk and eggs.

Cook-week job and schedule (per cycle)

Job	Amount
Cassava (1 block) / sweet potato from store	~45 kg / ~60 kg
Maize and cowpea from store	~25 kg each
Tilapia seined	~85 kg whole (~29 kg fillet)
Meat	~25 ducks; plus ~3–4 kids at the Feb and Aug cook weeks (7 a year)
Greens / tomato-pepper sauce	~30 kg / ~35 kg (tomato and pepper frozen whole earlier)
Eggs into meals	~550–700 per cycle; use egg-light recipes

Workload warning: ~420 kg of food in 6 days is heavy for 2–3 people. Use helpers or split the cook week into two parts. The house layout is built for it: harvest enters at the mudroom and flows through processing and freezers to the kitchen; the veranda is overflow workspace. Freezer, chill-room and power alarms (section 9) protect the week's output.

Storage

- Frozen cooked meals: 4 chest freezers of 300 L (1,200 L). One cycle needs ~550–600 L; the rest is backup and room for whole frozen tomato, pepper and fruit. Cooked meals keep best 3–4 months, so the 13-week cycle leaves little margin: seal flat, date everything. Each freezer carries a temperature probe on the alarm network.
- Dry: maize, cowpea, peanuts, sunflower seed, chia, honey, dried mushrooms in sealed jars. Cured: sweet potato (3–4 months), pumpkin (3–6 months). In the ground: cassava. Fridge: eggs (4–5 weeks), cheese, yogurt, kefir.
- Rule: harvest no more than fridge life, or cook and freeze it.

Year calendar (built from v3 sowing and breeding dates)

Month	Main work
Jan	Sweet potato, sunflower, pumpkin harvest; plant pumpkin; sow sunflower
Feb	COOK WEEK (kid slaughter); group B kids; sow maize, cowpea; set duck eggs
Mar	Sweet potato and peanut harvest; breed goat group A; sow cowpea
Apr	Pumpkin harvest; sow chia/perilla; finish peanuts
May	COOK WEEK; sunflower harvest; winter cover on beds
Jun	Sweet potato harvest; frost cloth; cover or move fish if below 20 °C
Jul	Vitamin D test; duck dawn light; kale beds; hooves
Aug	COOK WEEK (kid slaughter); group A kids; sow sunflower
Sep	Sow maize, cowpea; plant pumpkin; breed group B; lime and compost
Oct	Sow peanuts; heat checks begin; weekly FAMACHA
Nov	COOK WEEK; finish peanuts; shade and water for animals
Dec	Sunflower and pumpkin harvest; daily heat-stress checks

Quarterly (v5): review sensor calibration, test manual overrides and restock the spare-parts kit alongside the cook-week countdown. Calendar bar chart from v4 is represented here as a table.

8. Key weekly diet, animal feeding and cook schedule

One “portion” is a household serving of about 1.25 kg of cooked frozen food (two people). Four slots a day (breakfast, lunch, dinner, snack), ~5 kg a day. By energy the man eats about two thirds (~3.3 kg a day) and the woman one third (~1.7 kg). Fresh milk, eggs and the weekly basket come on top.

Day	Breakfast	Lunch	Dinner	Snack
Mon	Sopa paraguaya	Goat-bean-greens stew	Baked tilapia + sweet-potato mash	Chipa
Tue	Mushroom-egg tortilla	Duck locro	Tilapia + sweet-potato fish cakes	Seed bars
Wed	Sopa paraguaya	Duck or goat curry	Baked tilapia + sweet-potato mash	Chipa
Thu	Mushroom-egg tortilla	Goat-bean-greens stew	Tilapia + sweet-potato fish cakes	Seed bars
Fri	Sopa paraguaya	Duck locro	Baked tilapia + sweet-potato mash	Chipa
Sat	Mushroom-egg tortilla	Duck or goat curry	Tilapia + sweet-potato fish cakes	Seed bars
Sun	Mushroom-egg tortilla	Goat-bean-greens stew	Baked tilapia + sweet-potato mash	Chipa

Every day fresh: milk, yogurt or kefir ~2 L; eggs ~3–4; weekly basket (salad greens, herbs, scallion, moringa leaves, chillies, avocado, citrus, guava, banana). Man adds extra milk, seeds and honey if energy runs low. In week B swap sopa with tortilla and chipa with bars. Duck and pumpkin hash replaces a locro or curry once its portions are set. Per day each slot carries ~2,200 kcal and ~110 g protein across both people (estimate, ~20% error).

What the farm must deliver each week (household)

Food	Week	Cycle	Year	Food	Week	Cycle	Year
Cassava	3.4 kg	45 kg	179 kg	Eggs (need 2,300)	44	575	2,300
Sweet potato	4.6 kg	60 kg	241 kg	Milk, yogurt, kefir	14.0 L	182 L	730 L
Maize	2.0 kg	26 kg	102 kg	Goat cheese	0.21 kg	2.8 kg	11 kg
Cowpea	1.9 kg	25 kg	99 kg	Tilapia fillet	2.2 kg	29 kg	117 kg
Peanuts	0.42 kg	5.5 kg	22 kg	Meat (duck, goat)	2.5 kg	32 kg	128 kg
Sunflower seeds	0.42 kg	5.5 kg	22 kg	Greens, tomato, pepper	6.3 kg	82 kg	329 kg
Chia	0.29 kg	3.8 kg	15 kg	Pumpkin	2.2 kg	28 kg	113 kg
Honey	0.29 kg	3.8 kg	15 kg	Fruit	3.6 kg	46 kg	186 kg
Mushrooms	1.1 kg	13.8 kg	55 kg	Avocado flesh	1.1 kg	14.2 kg	57 kg

Animal feeding: what, how much, when

Animal	What they eat	Per day	Per week	Per year	Source and check
Goats (~8 adults + kids)	Napier, moringa, pigeon pea, mulberry, sweet-potato vines, wilted cassava leaf, crop residues; tannin browse; mineral block + salt + copper	~16 kg DM	~115 kg DM	~6,000 kg DM	Supply 7,800 kg (+30%). A 1,300 m² paddock at 1.2 kg DM/m²/yr gives ~150 kg DM per 35-day regrowth, above the 115 kg a week one paddock must supply.
Muscovy (22–27 adults + ~25 growing per cycle)	Dry maize, BSF larvae, duckweed, orchard and pasture forage, kitchen scraps, eggshell/bone meal	3.3–4.1 kg grain-eq.	23–29 kg	1,200–1,500 kg	Maize gives ~24 kg a week; BSF and duckweed must cover 0–8.7 kg a week (0–450 kg a year). Add ~600 m² maize if the BSF trial fails.
Tilapia + traíra (4 ponds)	Duckweed, BSF larvae, duck-manure pond fertility, second-grade roots, natural pond growth	~22 kg fresh duckweed	~154 kg	~8,000 kg	Weekly water and feeder check; cover or move fish below 20 °C.
Black soldier fly (~20 totes)	Fish frames and guts (~56 kg per cook week), peelings, spoiled produce, offal, scraps, manure	Twice a week	2 feedings	Larvae every 2 days	Output unknown until trialled; it decides the maize area.
Bees	Orchard, sunflower, farm flowers	-	-	-	Check for swarming and disease.

Feed loop: everything the animals eat is grown on the property; the only bought items are the mineral block and vet kit. Goat and duck manure, BSF frass and pond water go to compost and beds; bed runoff never goes to ponds.

Animal feeding week

Day	Goats	Ducks	Fish + BSF	Other
Every day	Milk, water, shade, heat check in summer	Collect eggs (every 1–2 days), clean water, ration grain (timed feeder + check), night lock-up	Collect larvae every 2 days	Egg count, milk volume log; dashboard check
Mon	Open the next paddock (1 of 5, ~7 days each)	Move duck runs	-	Weekly basket harvest
Tue	FAMACHA check (summer)	-	Feed BSF	-
Wed	Mineral feeder and salt check	Check nest boxes	Check pond water and feeders	Duckweed harvest for fish
Thu	Paddock walk: browse and regrowth	-	-	Compost turn
Fri	Hooves (every 3 months)	Clean house, fresh straw	Feed BSF	Spare-parts kit and sensor check (v5)
Sat	-	-	Collect and bank frass	Weekly basket harvest

Day	Goats	Ducks	Fish + BSF	Other
Sun	Milk and water only	Eggs and water only	Larvae	Rest

Cook schedule

Cook week	Pattern (13 weeks apart)	Meat	Also happening	Season note
Feb	Week 7	~25 ducks + 3–4 kids (born Aug–Sep, ~6 months)	Group B kids born; duck hatching; sow maize, cowpea	Hot, humid: keep the chain cold
May	Week 20	~25 ducks (no kids)	Sow chia; sunflower harvest; winter beds	Cooler, easier cooking days
Aug	Week 33	~25 ducks + 3–4 kids (born Feb–Mar)	Group A kids born; duck hatching; sow sunflower	Cool, dry, frost risk for tomato
Nov	Week 46	~25 ducks (no kids)	Peanut sowing; summer heat begins	Heat returns: shade and cold water

Four cycles are 91 days (13 weeks) apart, not 12 (which would give 4.3 cycles); raw goat from the Feb and Aug kills is frozen for the May and Nov meals.

Balanced seven-day cook week and crew

Day	Work (07:00–12:00 / 13:00–18:00)	Output	Crew (2–3)
1	Slaughter and butcher ducks and kids in the work barn; seine one pond and fillet tilapia / dig cassava, cut greens, thaw tomato-pepper, start bone broth, soak beans, waste to BSF	Prepped meat, ~29 kg fillet, broth on	Butcher, prep, runner
2	Strain broth; stew and locro bases / cook stew and locro; cool; bag flat	65 portions (~81 kg)	Cook, packer
3	Curries and tomato-pepper sauce (~35 kg) / chipa dough and bake; cool and bag	72 portions (~90 kg)	Cook, baker, packer
4	Sopa paraguaya bake / tortilla squares; cool and bag	91 portions (~114 kg)	Baker, packer
5	Baked tilapia in sauce / fish cakes; cool and bag	91 portions (~114 kg)	Cook, packer
6	Seed, honey and banana bars / vacuum seal, label, load freezers oldest at front, log inventory	45 portions (~56 kg)	Whole crew
7	Clean kitchen and processing room, waste to BSF and compost, bones to bone meal; rest	Inventory sheet filed	Rest day

Countdown around each cook week

When	Do
18 weeks before	Set Muscovy eggs under mothers (35 days to hatch, then 12–14 weeks growing) so the ducks are ready at this cook week
4 weeks before	Count freezer space (need ~550–600 L free of the 1,200 L), clean freezers, order salt, bags and vacuum rolls, check beans and dry stores; **v5:** test freezer alarms and phone notifications
1 week before	Fix helpers, sharpen knives, **check power, battery state, generator and ATS**, confirm which pond to seine, plan which kids go (Feb and Aug)
The day before	Fast the ducks overnight, thaw whole tomato and pepper, set up the mudroom-to-kitchen flow, pre-chill the processing room and confirm chill-room temperature log
Day 1 to 7	Follow the timetable; log weights and kg per dish; photograph the freezer layout
After	Record yields, egg counts and labour hours; update the nutrient balance and next cycle's areas; review dashboard fault history

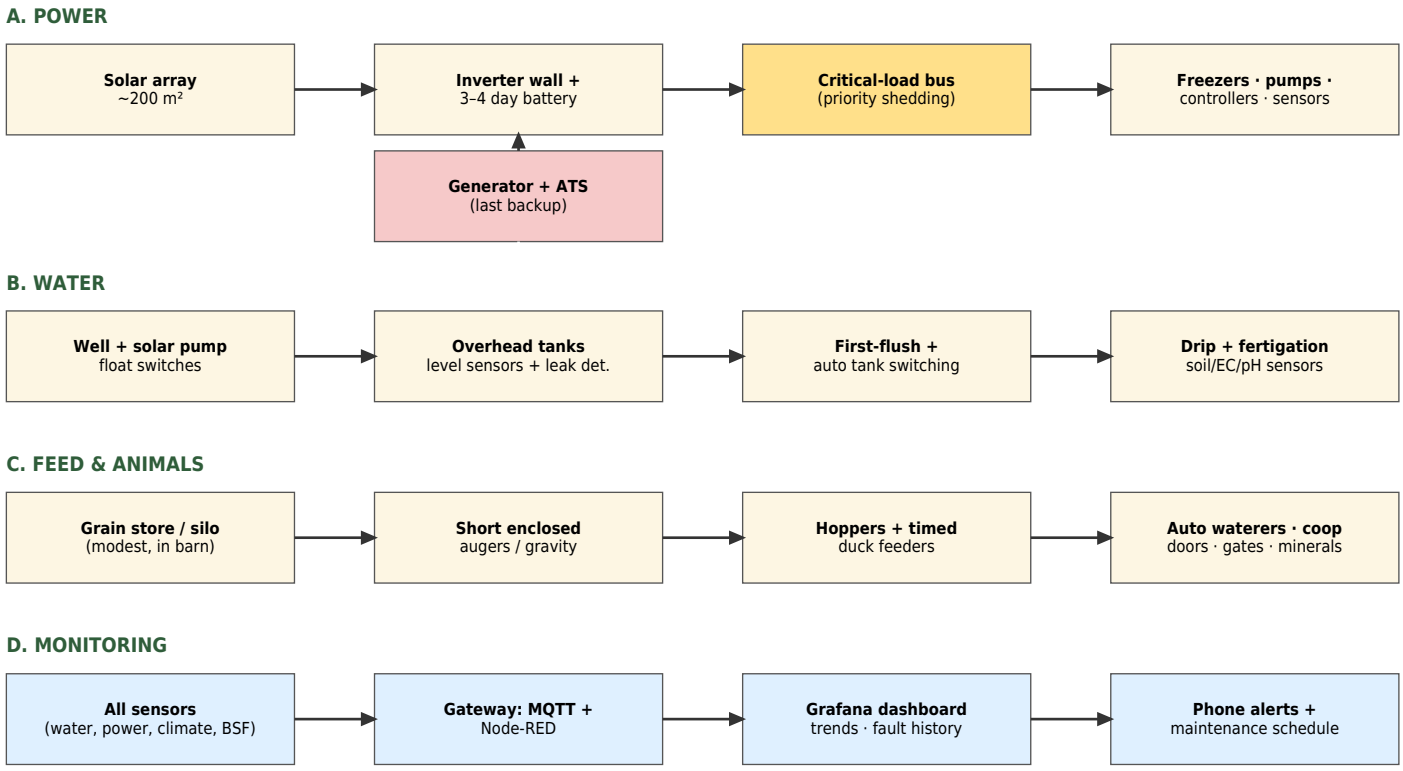
9. Automation layer (new in v5)

Principle: automation is a tool for reliability and labour reduction, not a goal in itself. Sensors + simple mechanical systems + a spare-parts kit beat complex robotics. Everything has a manual override; no single device may be able to starve, flood or freeze the farm.

Priority order (reliability first)

#	Layer	Why this rank
1	Water security	Every animal, bed and the house depend on it; peak need ~7 m³/day
2	Power security (freezers & critical loads)	~455 kg of cooked food per cycle sits in the freezers
3	Animal basics (feed, water, doors, alarms)	Daily, unforgiving chores; predator lock-up
4	Environmental control (greenhouse, mushroom room, irrigation)	Protects yields; slower failure
5	Monitoring & alerts	Makes layers 1–4 visible from the phone

Architecture diagram



Every loop keeps a manual override. Spare-parts kit + labelled workbench back all four layers.

Reliability order: water → power → animal basics → environment → monitoring

Automation architecture: solar → inverter → critical loads; silo → short augers → hoppers; sensors → dashboard → phone.

Device & method grid (only what suits 2 ha / 1 operator)

Area	Practical devices and methods	Failure fallback
Water & irrigation	Solar-powered well pump + float switches + overhead tanks; soil-moisture + EC/pH sensor network → drip irrigation + simple fertigation dosing pumps; weather-station integration; leak detection + tank-level alarms; first-flush diverter + automatic tank switching; rainwater harvesting (already in plan)	Manual valves on every branch; gravity-fed tank buffer of several days; spare pump and valves
Feed & animals	Modest central grain store + short enclosed augers or gravity hoppers (no long commercial flex-auger network); timed / gravity drop feeders for ducks; automatic waterers with level sensors; app- or timer-controlled electric gates for duck cells and goat paddocks; automatic coop doors; BSF self-harvesting totes with temperature/humidity sensors; feed-level sensors in hoppers; simple cameras (duck house, goat shed, BSF); automatic mineral feeders for goats	Gravity hoppers work without power; manual gate latches; coop-door manual lock; hand feeding routine written on the mudroom wall
Power & energy	Inverter wall + 3–4 day battery autonomy dedicated to critical loads; Automatic Transfer Switch (ATS) + generator; freezer temperature alarms; critical-load priority shedding	Generator started by hand; fuel stored; freezers kept shut
Climate & processing	Greenhouse / rain-shelter vents + temperature/humidity sensors; mushroom-room climate sensors; linear one-way cook-week flow (kept and monitored); chill-room alarm	Manual vent props; spare fans and misters
Plants (extra)	Soil sensors + fertigation; weather station; automated shade / frost-cloth on sensitive beds	Hand-placed cloth; timer fallback on drip
Animals (extra)	Livestock health / activity cameras; BSF environmental sensors	Daily walk-round remains the primary check
Monitoring & safety	Central dashboard (Node-RED / MQTT / Grafana or equivalent); phone notifications + trend analysis + fault history + maintenance schedules; isolation, redundancy and remote alerts; spare-parts kit + labelled workbench emphasised as more valuable than extra automation layers	Local alarms (sirens/lights) if the network drops; paper checklist

Explicitly excluded (over-engineered for this scale)

- Long chain-disk conveyors and long commercial flex-auger networks.
- Full robotic feeders.
- Complex AI pest systems.
- Automated egg collection (hand-collect every 1–2 days).

Spare-parts kit and workbench (mandatory)

Pumps (one spare of each type), valves, filters, sensors, irrigation fittings, belts, fuses, gate and door motors or actuators, cable and connectors. Labelled bins on the workshop bench in the work barn, inventory list on the dashboard, restocked each quarter. A spare-parts kit and labelled workbench are ranked above any extra automation layer: repairability is part of the labour strategy.

Automation failure modes and mitigation

Failure mode	Consequence	Mitigation
Sensor failure (drift, dead, wrong reading)	False alarm or missed alarm; over/under irrigation	Two sensors on critical points; calibration schedule; manual check routine; spares
Pump failure	No water to tanks or beds	Spare pump, overhead tank buffer, manual valves, rain tanks, generator
Network / dashboard outage	No remote alerts	Local alarms, loops continue on local controllers, paper checklist, manual override
Power failure	Freezer loss, no pumping	3–4 day battery on critical loads, ATS + generator, load shedding, freezer temperature alarm
Gate / door actuator fails	Predator exposure or stuck animals	Manual lock, camera check, spare actuator
Software or configuration error	Unwanted dosing or watering	Hard physical limits (float switches, timers), dosing pump caps, version-backed configuration

10. Labour and single-operator plan

Target: average ≤5–6 hours of farm work a day outside the four cook weeks (unverified: log actual hours in years 1–2). Daily: goat milking, animal checks, egg collection, water and feed. Weekly: move duck runs, BSF collection, paddock rotation, weeding, bed check. Quarterly: hooves, sensor and spare-parts review. Seasonal: planting, major harvests, infrastructure maintenance.

How the automation layer reduces daily chores (v5)

Chore	Before (v4)	With automation (v5)
Watering beds and orchard	Daily judgement and manual valves	Soil-moisture + weather-driven drip; only exceptions need a visit
Feeding ducks	Ration morning and evening by hand	Timed / gravity drop feeders; hopper level sensors; refill weekly
Goat water and minerals	Daily check	Automatic waterers with level sensors, mineral feeders; weekly check
Checking tanks and freezers	Daily walk or reading	Tank-level, leak and freezer temperature alarms to the phone
Rotating paddocks and duck cells	Open and close by hand	App- or timer-controlled gates; manual override
Locking up ducks	Hand lock at dusk	Automatic coop door + camera check
Greenhouse / shelters / mushroom room	Open vents, mist by judgement	Vent and climate sensors; automated shade / frost cloth

- Repairability and spare parts are part of the labour strategy: a failed pump with no spare costs more hours than the automation ever saved.
- Expected saving is a planning assumption only; log hours for two years and re-set the target from data.

If the operator is sick or hurt for 1–2 weeks (proposal)

- Written daily checklist (milking, eggs, water, freezer alarm) on the mudroom wall; a trusted neighbour lined up for goat milking (the buck-swap relationship is the natural contact).
- Automation covers water, feeders, doors and alarms; frozen stock is the food buffer; generator + ATS + freezer alarm cover power failure; the dashboard can be shared with the neighbour.
- Schedule no slaughter or cook week within a week of travel; dry off does earlier if an absence is planned.

11. Water, power and sanitation

Water (hard gate). Peak demand ~7 m³/day in the driest month. Confirm rights with MADES, test well yield under sustained pumping, size rain tanks for the dry-season deficit, use drip so bed runoff never reaches ponds. Write a monthly balance and verify it before earthworks. The well and pump house (3 × 3 m) sit south of the compound beside the solar strip (v5.2); final position after the yield test; keep at least 30 m from the duck house and goat shed and ~50 m from compost and humanure. **v5 additions:** leak detection, tank-level monitoring and alarms, first-flush diverter with automatic tank switching, float switches on the overhead tanks.

Use	Driver (from plan)	Jan–Dec value
Irrigation	~2,650 m² food + 1,800 m² maize + forage	TBD; peak month ~7 m³/day total
Animals	6–8 goats, 22–27 ducks + grow-out, BSF	TBD
Ponds	4 × 500 m² top-up and evaporation	TBD
Lap pool (v5.2)	112.5 m² surface: ~150 m³ first fill, ~0.5–0.7 m³/day evaporation in summer (estimate), cover	TBD; add to peak-month check
House + processing	2–3 people, cook weeks	TBD
Supply	Well yield + rain-tank volume	TBD (test first)

Power budget re-run (v5 estimate: measure real loads once installed)

Freezers ~1,000 kWh a year (~2.7 kWh/day) must stay cold. The v4 estimate of 8–11 kWh usable battery covered the freezers alone. v5 adds the automation loads below. The sensor/controller/camera numbers are rough proposals to be replaced by measurements.

Critical load (3–4 day autonomy)	Estimate	Notes
4 chest freezers	~2.7 kWh/day	From v4
Controllers, gateway, sensors, cameras	~0.7–1.4 kWh/day (30–60 W continuous)	Estimate; measure
Auto coop door, gates, feeders, waterers	Small, intermittent	Estimate; measure
Greenhouse / shelter vent fans, mushroom-room climate gear	Intermittent	Size after the first season
Chill room (work barn) at cook week	Run on battery (v4 rule)	Peak load; generator may assist
Well / irrigation pump	Solar-direct plus tank buffer	Not on the battery critical bus; tanks cover several days
Pool pump + filtration (v5.2)	~3 kWh/day (0.5 kW × ~6 h), estimate	Solar-direct in daytime, non-critical bus, first load shed
Indicative critical bus total	~3.5–4.5 kWh/day → ~11–18 kWh usable battery for 3–4 days	Before lights, tools, vacuum sealer; add margin; re-size array after measurement

Critical-load priority (shedding order): freezers and freezer alarm → controllers and network → water pump / tank valves → animal doors, gates, feeders → greenhouse vents and mushroom-room gear → chill room (except at cook week) → comfort loads. A small generator with ATS is the last backup. Load list to measure: 4 chest freezers, well/irrigation pump, lights and communications, workshop tools, vacuum sealer, cook-week kitchen loads, BSF and mushroom-room fans, goat-shed and pond aeration if used, plus all automation loads.

Sanitation and humanure. A composting toilet or managed humanure system closes the fertility loop: low odour, easy handling and proven pathogen-kill time before use on food crops. Proposal (verify with extension/SENACSA): hot-compost, then rest at least 12 months; use first on orchard trees, duck maize and goat forage; not on salad, greens or root crops eaten raw until the system is tested. Keep wells 30 m+ from animals and humanure.

12. Capital, external inputs and build order

External by design: salt and iodine; a small vet and vaccine kit and a copper mineral block; mushroom spawn (until you make your own); start-up items (foundation stock, solar, lime, rock phosphate); algae oil, and vitamin D in winter if blood tests are low. Everything else, including all animal feed, is grown on the property.

Phase	Build	Spend categories	Cost
0–6 months	Water rights, well and dam test, boundary survey, road access; confirm local availability of sensors, valves and Node-RED-compatible hardware	Legal fees, surveys, well test	TBD
0–9 months	Rainwater tanks, secure core; first spare-parts kit	Tanks, fencing, first compost system	TBD
6–18 months	House core (191 m²), work barn, well and pump, rain shelters, first staples and orchard, compost, sanitation; drip + soil sensors; controller / dashboard head-end	House, shelters, drip, seedlings, sensors, controllers	TBD
12–24 months	BSF, Muscovy ducks, duck pasture; auto coop door, feeders, gates	Sheds, stock, fencing, doors, actuators	TBD
18–30 months	Goats, ponds, bees, mushroom room, power system finalised (inverter wall, battery, ATS, generator)	Stock, ponds, freezers, solar, battery, ATS	TBD
24–36 months	Full system, activity annex (music room, gym) and lap pool + plant room if not built earlier, first complete cook weeks, data for nutrient re-run	Annex, generator, helpers, amendments	TBD

Updated capital / TBD list (v5 additions)

- Sensors: soil moisture, EC/pH, temperature/humidity, tank level, leak, freezer probes.
- Controllers and gateway (Node-RED / MQTT / Grafana host), network hardware, cameras.
- Auto coop doors, electric gates, timed or gravity feeders, automatic waterers and mineral feeders.
- Automatic transfer switch, critical-load panel, extra battery capacity for automation loads.
- Spare-parts kit and labelled workbench stock.
- Private-zone buffer planting and garden landscaping; showpiece beds and potting room fit-out; dog yard fencing; sound isolation for the music room and gym annex; lap pool with cover, filtration and plant room; pool fence, gate and alarm.

Proposal: hold a contingency reserve of 15–20% of build cost plus one year of running costs. Quote real prices before committing; none are invented here.

13. Year 1–2 trials scoreboard

Trial	Pass line	If it fails
Well	≥7 m³/day in the driest month	More tank storage, or stop
Double sowing	Both sowings reach 0.59 / 0.37 / 0.077 kg/m² a year (maize / cowpea / sunflower)	Use the expansion reserve; double those areas
Tomato + pepper beds	≥3.2 kg/m²	Enlarge or add shelters
Greens / salad beds	≥1.8 / ≥3 kg/m²	Enlarge beds
Chia	≥0.1 kg/m²	Switch to perilla, then algae oil
Mushrooms	~1 kg per bag-cycle	More bags, or buy spawn
Duck grain	Maize 0.7 kg/m²: 1,050–1,260 kg vs need 1,200–1,500; BSF + duckweed cover 0–450 kg	Add ~600 m² maize
Eggs	≥2,300 a year	Khaki fallback
Milk	~1,100 L a year, ≥3 does always in milk	Add a doe
Labour	≤5–6 h/day logged	Cut scope or hire help
v5: Sensor + dashboard reliability	Alerts reach the phone; no unexplained sensor dropout over a full season; manual overrides tested each quarter	Simplify, replace sensors, add local alarms
v5: Automation labour saving	Logged hours fall measurably versus the manual routine	Remove automations that cost more upkeep than they save

14. Hard rules, risks and legal checklist

Hard rules

1. Every nutrient has at least two food sources where possible.
2. Re-run the nutrient balance whenever species or counts change (especially after Muscovy egg data).
3. Keep ~30% surplus on every crop and animal food (eggs: see decision).
4. Harvest no more than fridge life, or cook and freeze immediately.
5. Trial any new species on 20 m² before scaling (chia, tomato/pepper in soil beds, mushrooms, BSF output).
6. Never send bed runoff into the fish ponds.
7. Cover or move fish if water falls below 20 °C.
8. Check goats daily for heat stress in summer.
9. Increase soil organic matter every year.
10. Legal checks before building, slaughtering, selling or digging a dam.
11. Confirm double sowing of maize/cowpea/sunflower or double those three areas.
12. Trial BSF output before finalising the maize area.
13. **v5:** Every automated function keeps a manual override.
14. **v5:** Keep the spare-parts kit complete; it outranks adding automation.
15. **v5.2:** Pool water, backwash and chlorine never reach ponds, beds or the well.

Open risks

Risk	Mitigation
Double sowing unreliable	Expansion reserve; double the three areas
Egg shortfall (Muscovy)	Egg-light recipes; Khaki fallback; re-run balance
Soil-bed yields guessed; chia untested; ALA and vitamin D gaps	20 m² trials; perilla; algae oil; bloodwork
Duck grain tight; sandy soil years 1–3	BSF trial, maize buffer; compost, biochar, cover crops, lime + rock phosphate
MADES may restrict tilapia; growth stops below ~20 °C	Check rules first; deep covered pond; harvest by weight
Water rights and well yield (~7 m³/day peak)	Confirm with MADES and test before building
Goat heat stress and parasites	Shade, rotation, FAMACHA checks
Mushroom contamination	Clean room, pasteurised substrate, small batches
Winter frost and cold	Frost cloth (automated shade/frost-cloth), rain shelters, frost-tolerant fruit
High iron, vitamin A or selenium for the man	Ferritin test once; no liver; avoid extra supplements with these
Single-operator illness or injury	Contingency on the labour page; automation + neighbour
Freezer or power failure	3–4 days of battery on critical loads, generator + ATS, freezer temperature alarm, load shedding
v5: Automation failure (sensor, pump, network outage)	Spare parts + manual override + 3–4 day battery; local alarms; paper checklist (see section 9)
v5: Local availability of simple sensors, valves and Node-RED-compatible hardware unconfirmed	Confirm suppliers in months 0–6; choose widely available, repairable parts; keep spares
v5.1 / v5.2: Re-cut site is still a concept; active area grew to ~1.66 ha	Survey first; phase the build (house core, showpiece garden, staples first); defer the outer belts and reserve cropping if labour runs high
v5.2: Pool adds water, power and upkeep; safety with dogs and when swimming alone	Cover, fence + self-closing gate + alarm, solar-direct pump on the non-critical bus, phone check-in when swimming alone, drain away from ponds, beds and the well

Legal checklist

- Municipalidad de Villarrica: zoning.
- MADES: water, dam, well, fish species.
- SENACSA: slaughter (work barn), dairy, animal health.
- INFONA: forestry and land use.
- Local extension office: bees, composting, burn permits, food safety, vaccines.

15. Closing notes

- All yields remain untested planning estimates; re-run the nutrient balance before building.
- The first two years are for proving yields, refining labour, building soil and shaking down the automation. Only after that data exists should the system be considered truly self-sustaining.
- Philosophy (v5.2): two layers. The private compound is for daily enjoyment (nature, training, swimming, screens, music, food, rest); the productive land is for reliability, nutrition, low labour and closed loops. Life-critical systems win any conflict.
- Automation is a tool for reliability and labour reduction, not a goal in itself.
- v5.1: the site plan and house are a concept re-cut around the resident's hobbies and a real private zone; confirm wind, slope, boundary and well yield on the survey, then redraw scaled plans.